

Operation Research (OR)

Question :-

Imp Question :-

- Any Question convert in linear - Programming model (LPM) → 2 Question
- we use the solution of that inequality which z is maximum.

Important definition :-

- i) operation research
- ii) characteristic
- iii) phases of an or study
- iv) decision variables

Concept:-

Its linear Programming

linear inequality

cost → mini

profit → max

Question Method:-

- First, create Table and gain profit/cost
- Second, objective function
how much — ? let x_1 and x_2 variable
and z is profit. z is objective function

- Third, conditions
- Then Subject to constraints
- use restriction for conditions
- last Decision variable
(positivity of decision variable
non-negativity " ")

Solution To LP-model:-

- ↓
 - Graphical (where There are only two decision variable)
 - Simplex method
- ↓

Graphical Method

Question # 01 :-

Reddy mikks produce both interior and exterior paint from two raw material M_1 and M_2 . The following Table provides, the basic data of the problem.

Pounds of Raw material per Tons of	Exterior Paint	Interior Paint	Maximum daily Tons
Raw material M_1	6	4	24
Raw material M_2	1	2	6
Profit Per Ton	5	4	

₹ 1000

A market Survey indicates that, the daily demand for interior paint can not exceed that of exterior paint by more than 1. Ton. Also the maximum daily demand of interior paint is 2 Tons. Reddy mikks wants to determine the optimum (best) product mix of interior and exterior paints that maximizes the Total daily profit.

Solutions-

let x_1 and x_2 be Tons Produced daily of exterior and interior paint of exterior and interior paint. let z be the objective function. Then the mathematical model of reddy mikks problem is as follows minimize

$$z = 5x_1 + 4x_2$$

Subject To:-

$$L_1 : 6x_1 + 4x_2 \leq 24 \quad \text{--- (i)}$$

$$L_2 : x_1 + 2x_2 \leq 6 \quad \text{--- (ii)}$$

$$L_3 : x_2 \leq x_1 + 1 \Rightarrow x_2 - x_1 \leq 1 \quad \text{--- (iii)}$$

$$L_4 : x_2 \leq 2 \quad \text{--- (iv)}$$

$(x_1, x_2 \geq 0)$

Graphical Solution is as:

Step-1 make the constraint equation

$$6x_1 + 4x_2 \leq 24$$

$$3x_1 + 2x_2 \leq 12 \quad \text{--- } L_1$$

$$x_1 + 2x_2 = 6 \quad \text{--- } L_2$$

$$-x_1 + x_2 = 1 \quad \text{--- } L_3$$

$$x_2 = 2 \quad \text{--- } L_4$$

Step-II Find coordinates of each method

when $x_1 = 0$ and then $x_2 = 0$

For L_1 $3x_1 + 2x_2 = 12$

$x_1 = 0$ then $x_2 = 0$

$$\begin{cases} x_1 = 0 & x_2 = 6 \\ x_1 = 4 & x_2 = 0 \end{cases}$$

For L_2 $x_1 + 2x_2 = 6$

$$\begin{cases} x_1 = 0 & x_2 = 3 \\ x_1 = 6 & x_2 = 0 \end{cases}$$

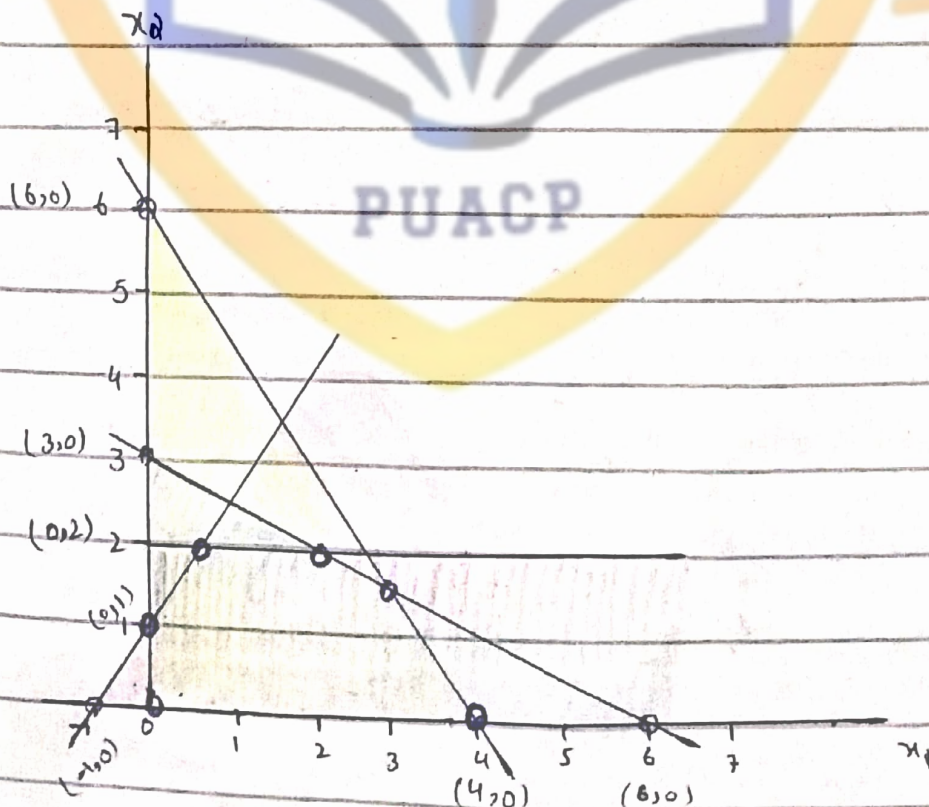
For L_3 $-x_1 + x_2 = 1$

$$\begin{cases} x_1 = 0 & x_2 = 1 \\ x_1 = -1 & x_2 = 0 \end{cases}$$

For L_4 $x_2 = 2$

$x_1 = 0$ $x_2 = 2$

Step - iii make graph:



Step-iv Solution:-

$$\begin{array}{cccc} A(0,0) & B(0,1) & C(1,2) & D(2,2) \\ E(3,1.5) & F(4,0) & & \end{array}$$

Corner Point	$5x_1 + 4x_2 =$	Z
$A(0,0)$	$5(0) + 4(0) =$	0
$B(0,1)$	$5(0) + 4(1) =$	4
$C(1,2)$	$5(1) + 4(2) =$	13
$D(2,2)$	$5(2) + 4(2) =$	18
$E(3,1.5)$	$5(3) + 4(1.5) =$	21
$F(4,0)$	$5(4) + 4(0) =$	20

Step-v Conclusion:

If 3 Tons of exterior and 1.5 Tons of interior. Then will be max corn allow \$ Profit.

Problems- 2

A Company manufactures Two types of product/Products A and B and sells them at a profit of Rs. 4 on type A and Rs. 5 on B. Each Product is Processed on Two machines X and Y. Type A requires 2 mint of Processing Time on X and 3 mint on Y. and Type B requires 2 mint of processing Time on X and 2 mint on Y.

The machine X is available for not more than 5 hours 30 minutes while Y is available for 8 hours during any working day Formulate the Problem as a LP model and solve.

How much Production of both Types should the factory produce, to get max Profit.

Solution :-

∴ Table :-

	X	Y
A	2	3
B	2	2
	330	480

Objective function (Decision variable)

Let x_1 and x_2 be the number of products of Type A and B respectively. Then the mathematical model of the given LP is as under

maximum

$$z = 4x_1 + 5x_2$$

Subject To

$$2x_1 + 3x_2 \leq 330 \quad \text{--- i}$$

$$2x_1 + 2x_2 \leq 480 \quad \text{--- ii}$$

$$(x_1, x_2 \geq 0)$$

Step - 1 make the constraint equation

$$2x_1 + 3x_2 = 330$$

$$x_1 + x_2 = 165 \quad \text{--- } l_1$$

$$3x_1 + 2x_2 = 480 \quad \text{--- } l_2$$

Step II Find Condition of each method.

FOR L_1

$$x_1 + x_2 = 165$$

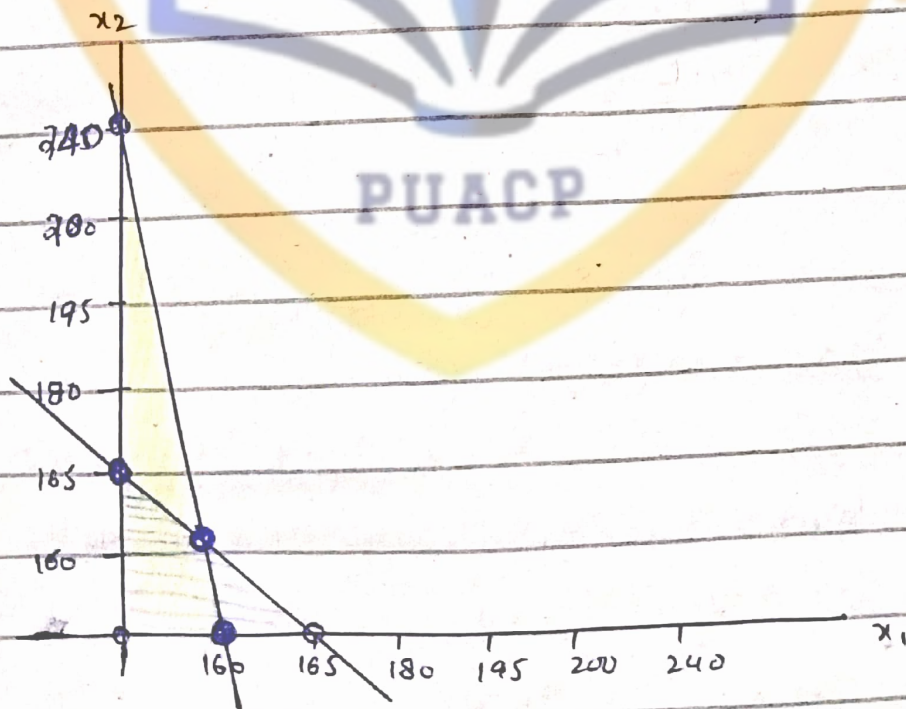
$$\begin{cases} x_1 = 0 & x_2 = 165 \\ x_2 = 0 & x_1 = 165 \end{cases}$$

FOR L_2

$$3x_1 + 2x_2 = 480$$

$$\begin{cases} x_1 = 0 & x_2 = 240 \\ x_2 = 0 & x_1 = 160 \end{cases}$$

Step - III Graph.



Step-iv Solution

$$A(0, 165)$$

$$B(150, 15)$$

$$C(0, 160)$$

$$2x_1 + x_2 = 165$$

$$2x_1 + 2x_2 = 330$$

$$-3x_1 + 2x_2 = 480$$

$$+x_1 = 150$$

$$150 + x_2 = 165$$

$$x_2 = 165 - 150$$

$$x_2 = 15$$

$$x_1 = 150$$

Corner

Point	$4x_1 + 5x_2 = Z$
$A(0, 165)$	$4(0) + 5(165) = 825$
$B(150, 15)$	$4(150) + 5(15) = 675$
$C(160, 0)$	$4(160) + 5(0) = 640$

Step-v Conclusion

So, $(0, 165)$ product is maximum, with maximum value (825) we should produce 165 products of Type B only.

Problem- 3

A person require 10, 12, 12 unit of chemical A, B, C respectively. A liquid product contains 5, 2 and 1 units of chemical A, B and C. respectively Per Tons. A dry Product contains 1, 2 and 4 units of A, B, C. respectively per carton. If the liquid Product cost \$ 3 Per Ton and dry product cost \$ 2 Per carton.

How many of each should be purchased to minimize the cost and meet the requirement.

Solution:-

First we put given information in the Table as below

	units/Ton	unit/carton	unit require
A	5	1	10
B	2	2	12
C	1	4	12

let x and y be the number of tons and number of carton that are purchased.

The mathematical model of the product is
and under minimize

$$z = 4x_1 + 5x_2$$

Subject To

$$l_1 : 2x_1 + 2x_2 \leq 330$$

$$l_2 : 3x_1 + 2x_2 \leq 480$$

$$z = 3x + 2y$$

Subject To

$$l_1 : 5x + y \geq 10$$

$$l_2 : 2x + 2y \geq 12 \rightarrow x + y \geq 6$$

$$l_3 : x + 4y \geq 12$$

$$(x, y \geq 0)$$

Graphical Method

Step 1 - make the constraints equation

$$5x + y = 10 \quad \text{--- } l_1$$

$$x + y = 6 \quad \text{--- } l_2$$

$$x + 4y = 12 \quad \text{--- } l_3$$

Step-II Find coordinate of each equation

For l_1 :

$$\left\{ \begin{array}{l} x = 0 \quad y = 10 \\ y = 0 \quad x = 2 \end{array} \right.$$

For L_2 :

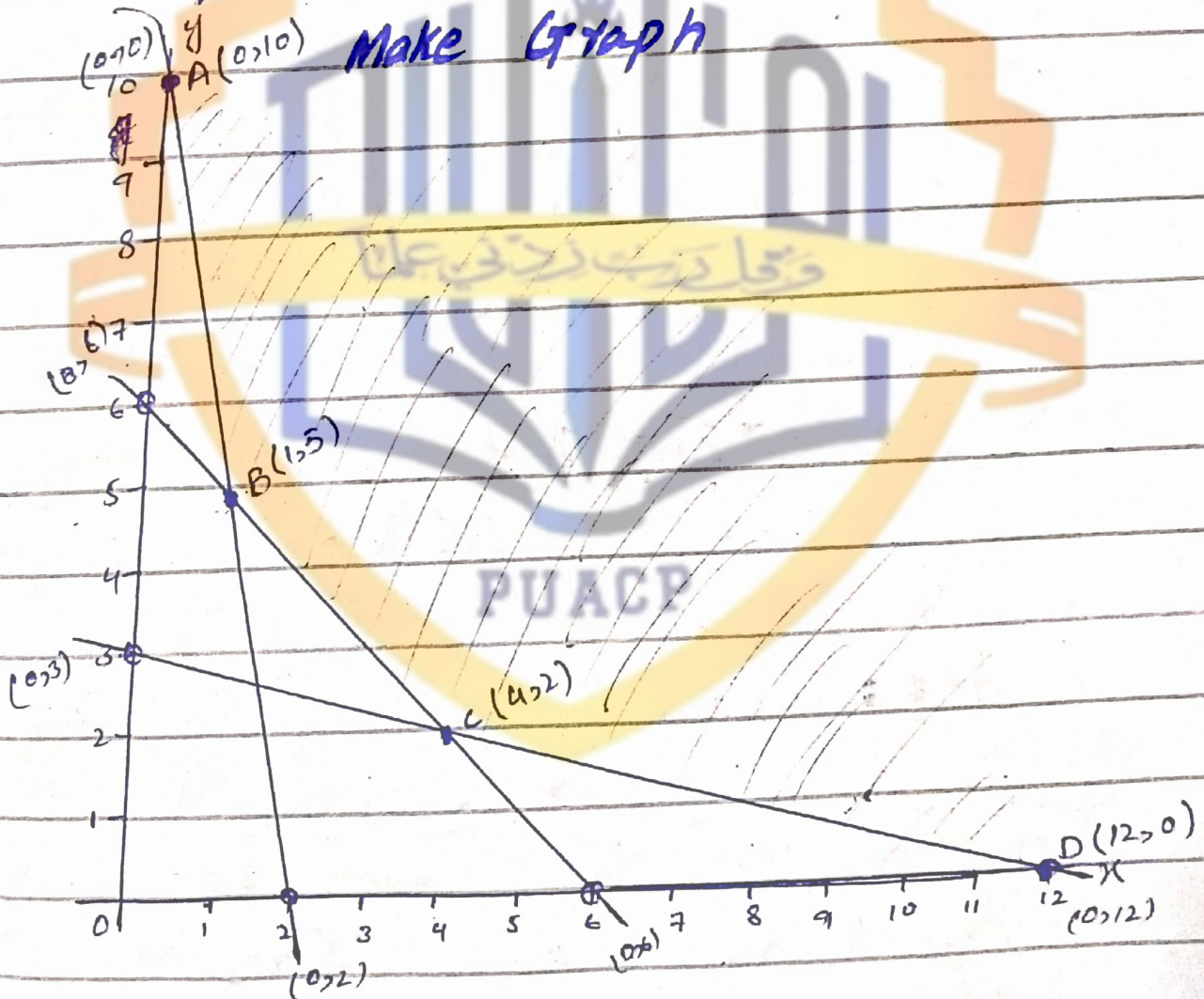
$$\begin{cases} x = 0 & y = 6 \\ y = 0 & x = 6 \end{cases}$$

For L_3 :

$$\begin{cases} x = 0 & y = 3 \\ y = 0 & x = 12 \end{cases}$$

Step-III

Make Graph



Step - iv

Solution

Corner

$$Z = 3x + 2y$$

A(0,10)

$$3(0) + 2(10) = 20$$

B(1,5)

$$3(1) + 2(5) = \underline{13}$$

C(4,2)

$$3(4) + 2(2) = 16$$

D(12,0)

$$3(12) + 2(0) = 36$$

Step - v Conclusion

Z is minimum at B(1,5) with minimum value 13.

Hence 1 Jar and 5 carton should be purchased to give the minimum cost of \$13.

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Problem - 4

Three products are produced to three different operations. The time required per unit per unit of each product, the daily capacity of the operation under profit per unit sold of each product are as.

operation	Prod-I	Prod-II	Prod-III	capacity
1	1	2	1	430
2	3	0	2	460
3	2	4	0	420
profit/unit	3	2	6	

Formulate the model to determine the optimum daily production for the three products that maximize the profit.

Solution :-

Let x, y, z be the number of units produced of product 1, 2 and 3 respectively. Then the mathematical model of the problem is.

$$\text{maximize } z = 3x + 2y + 6z$$

Subject To:-

$$x + 2y + z \leq 430$$

$$3x + 0y + 2z \leq 460$$

$$2x + 4y + 0z \leq 420$$

$$(x, y, z \geq 0)$$

Graphical Method

Step-i make the constraint equation

$$x + 2y + z = 430 \quad \text{--- } L_1$$

$$3x + 2z = 460 \quad \text{--- } L_2$$

$$2x + 4y = 420 \quad \text{--- } L_3$$

Step-ii Find coordinate of each equation

For L_1 :

$$\left. \begin{array}{l} x = 0 \\ y = 0 \end{array} \right\} \text{ not possible}$$

^

Problem-5

The required daily batch of feed mix for a broiler diet is 100 pounds. The diet must contain

-1 At least 0.8% but not more than 1.2% calcium

-2 At least 22% protein

-3 At least 5% fiber

The main ingredients used include limestone, corn and soyabean meal. The nutritive contents are.

ingredient	calcium	Protein	Fiber	cost
Limestone	0.380	0.00	0.00	0.0164
Corn	0.001	0.09	0.02	0.0645
Soyabean	0.002	0.50	0.08	0.1250

Formulate Linear programming model for determining feed mix which can be prepared at least cost.

Solution:-

let x, y, z denote the amount of line-stone, corn, Soyabean meal respectively.

Then the mathematical ~~no~~ L-P model is
minimize $z = 0.0164x + 0.0645y + 0.1250z$

Subject To:-

$$0.380x + 0.001y + 0.002z \geq 0.8$$

$$0.380x + 0.001y + 0.002z \leq 1.2$$

$$0.09y + 0.50z \geq 22$$

$$0.02y + 0.08z \geq 5$$

$$(x, y, z \geq 0)$$

Result:-

Three Dimension variable is not possible in Graphical method. Because Graphical method has one ~~one~~ or Two Dimension variable.

Problem - 6

A bank has Two Types of Branches A satellite branch employs branches: 3 people, requires \$100,000 to construct and open and generates an average daily revenue of \$10,000

A full service branch employs 6 people, requires \$140,000 to construct and open and generates an average daily revenue of \$18,000.

The bank has set up \$2.98 million available to open new branches and has decided to limit the new branches to a maximum of 25. If the bank further decides to hire at most 120 new employees.

How many branches of each type should the bank open in order to maximize the average daily revenue.

Solution :-

Let x and y denote the number of satellite and full service branches

respectively. The mathematical model of The program is as below

maximize :-

$$Z = 10,000x + 18,000y$$

Subject To

$$L_1 : x + y \leq 25$$

$$L_2 : 3x + 6y \leq 120 \Rightarrow x + 2y \leq 40$$

$$L_3 : 140,000y + 100,000x \leq 29,800,000$$

$$5x + 7y \leq 149$$

$$(x, y \geq 0)$$

Graphical method

Step - I make the constraints equation

$$x + y = 25$$

$$3x + 6y = 120$$

$$5x + 7y = 149$$

Step - II Find coordinate of each constraints

For L_1 :-

$$\left\{ \begin{array}{l} x = 0 \quad y = 25 \\ y = 0 \quad x = 25 \end{array} \right.$$

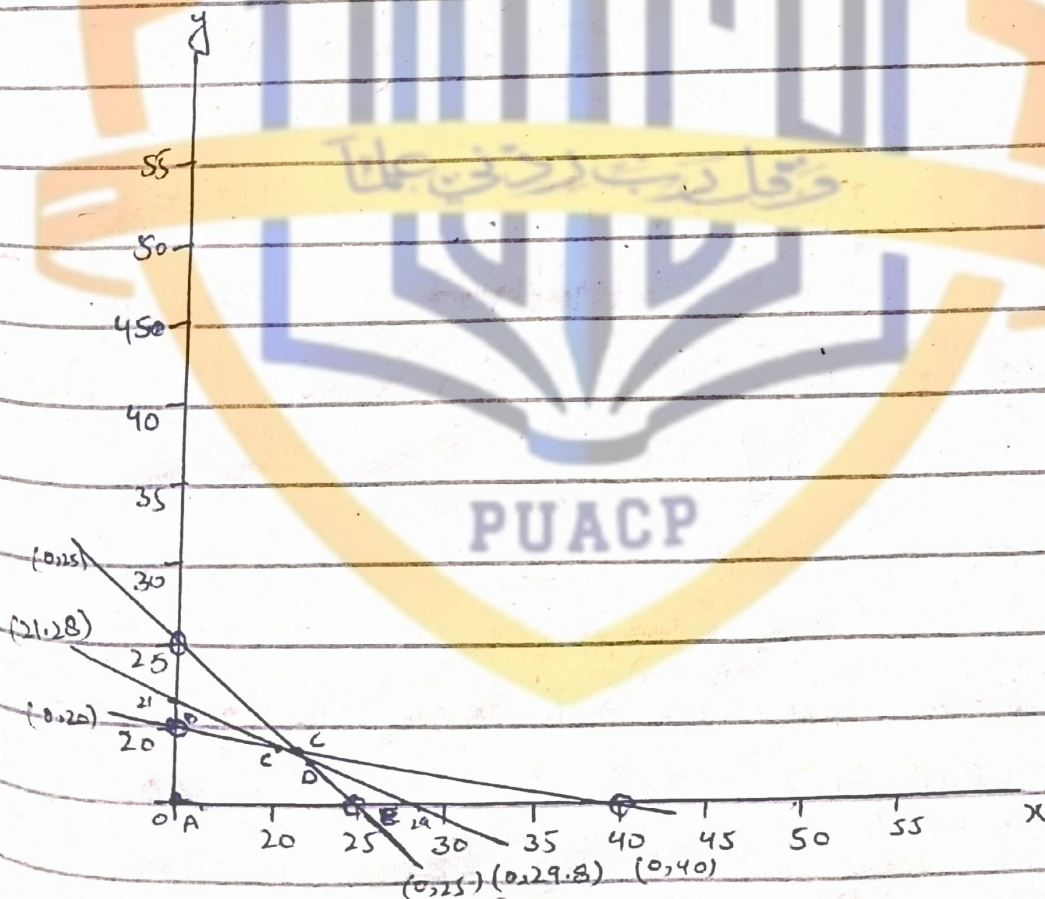
For L_2

$$\begin{cases} x=0 & y=20 \\ y=0 & x=40 \end{cases}$$

For L_3

$$\begin{cases} x=0 & y=21.28 \\ y=0 & x=29.8 \end{cases}$$

Step - III make Graph



Step-iv Solution

A(0,0)

B(0,20)

C(6,17)

D(13,12)

E(25,0)

corner point

$$Z = 10000x + 18000y$$

A(0,0)

$$10000(0) + 18000(0) = 0$$

B(0,20)

$$10000(0) + 18000(20) = 360000$$

C(6,17)

$$10000(6) + 18000(17) = 366000$$

D(13,12)

$$10000(13) + 18000(12) = 346000$$

E(25,0)

$$10000(25) + 18000(0) = 250000$$

Step-v

Conclusion

Z is maximum at C(6,17) and

maximum value is 366000, so we conclude

that 6 satellite branches and 17 full time

service branches should the bank open

to maximize the average daily revenue.

Problem - 7

A lake contains Two Types of fish I and II. The owner provides Two type of food A and B for These fish. Species I requires 2 units of food A and 4 unit of food B. Species II requires 5 units of food A and 2 units of food B. If The owner has 800 units of each food Find The maximum number of fish That The lack can support.

Solution

let x and y be number of fish of Type I and II respectively. Then The mathematical model of The problem is as

maximize $Z = x + y$

Subjct To

$$L_1 : 2x + 5y \leq 800$$

$$L_2 : 4x + 2y \leq 800$$

$$2x + y \leq 400$$

$$x, y \geq 0$$

Step-1 make constraint equation

$$2x + 5y = 800$$

$$2x + y = 400$$

Step-II Find coordinate of each constraint

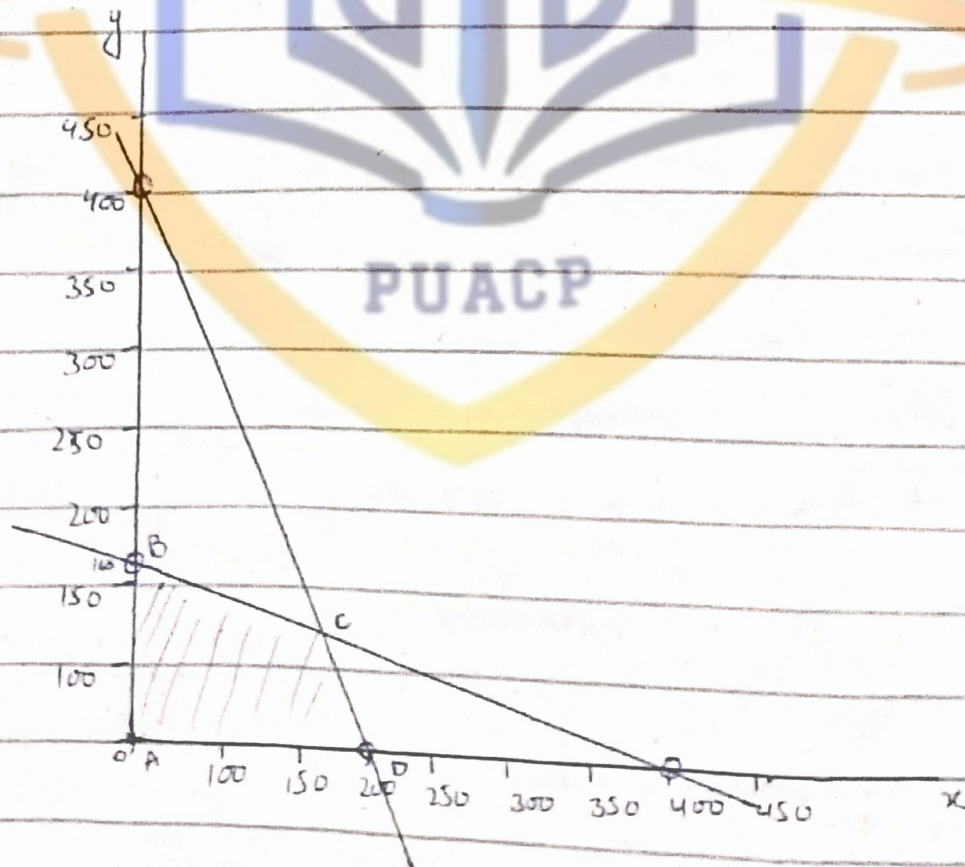
For L_1

$$\begin{cases} x=0 & y=160 \\ y=0 & x=400 \end{cases}$$

For L_2

$$\begin{cases} x=0 & y=400 \\ y=0 & x=200 \end{cases}$$

Step-III make Graph



Step-iv Solution

$$A(0,0) \quad B(0,160) \quad C(150,100)$$

$$D(200,0)$$

Corner point

$$x + y = z$$

$$A(0,0)$$

$$0 + 0 = 0$$

$$B(0,160)$$

$$0 + 160 = 160$$

$$C(150,100)$$

$$150 + 100 = 250$$

$$D(200,0)$$

$$200 + 0 = 200$$

Step-v

Conclusion

Hence z is maximum at $B(150,100)$ with maximum value 250.

So The number of fish of Type I are 150 and 100 of second type. That the lake can support.

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Problem - 8

A candidate wishes to use a combination of Radio and TV advertisement in his campaign. Research has shown that each one minute spot on TV reaches 0.09 million people and each one spot on Radio reaches 0.006 million. The candidate feels that he must reach at least 2.16 million people and he must buy a total of at least 80 minutes of advertisement.

How many minutes of each type should be used to minimize the radio cost \$100 per minute and TV cost \$500 per minute.

Solution :-

Let x and y be the number of minutes of Radio and TV bought. Then the mathematical LP model of the problem is as below

minimize:

$$Z = 100x + 500y$$

Subject To

$$J_1 = 0.006x + 0.09y \geq 2.16$$

$$x + 15y \geq 360$$

$$J_2 = x + y \geq 80$$

Step-I make The constraint equation

$$x + 15y = 360$$

$$x + y = 80$$

Step-II Find The coordinate of each constraint

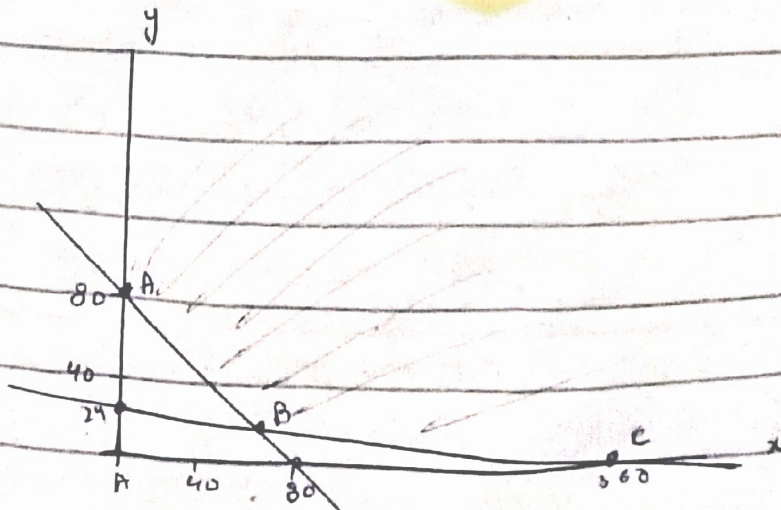
For J_1

$$\left. \begin{array}{l} x = 0 \\ y = 0 \end{array} \right\} \begin{array}{l} y = 24 \\ x = 360 \end{array}$$

For J_2

$$\left. \begin{array}{l} x = 0 \\ y = 0 \end{array} \right\} \begin{array}{l} y = 80 \\ x = 80 \end{array}$$

Step-III Graphs



Step-iv

Solution

A(0, 80)

B(60, 20)

C(360, 0)

corner point

$$Z = 100x + 500y$$

A(0, 80)

$$100(0) + 500(80) = 40000$$

B(60, 20)

$$100(60) + 500(20) = 16000$$

C(360, 0)

$$100(360) + 500(0) = 36000$$

Step-v

Conclusion

Z is minimum at point B(60, 20) with minimum value 16000. Hence we conclude that 60 minutes of Radio and 20 minutes of TV should be bought to pay the minimum cost \$16000.

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Problem no 9

An individual wishes to invest \$ 5000 over the next year in two types of investment. Investment A yields 5% and investment B yields 8%. market research recommends an allocation of at least 25% in A and at most 50% in B. moreover investment in A should be at least half the investment in B. How should the fund be allocated to the two investments.

Solution:-

let x and y represent investment A and investment B. Then the mathematical LP model of problem is as below

$$Z = 0.05x + 0.08y$$

$$d_1 : x + y \leq 5000$$

$$d_2 : x \geq 1250$$

$$d_3 : y \leq 2500$$

$$d_4 : x \geq \frac{1}{2}y$$

$$x - 0.5y \geq 0$$

$$(x, y \geq 0)$$